American Computer Science League

2025-2026 ● Contest 4: Short Problems Solutions ● Classroom Division

1. Graph Theory

Squaring the adjacency matrix for paths of length 2 yields the following:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

There are 13 entries of 1.

E. 13

2. Graph Theory

By inspection, there are 2 cycles in this directed graph. They are: ABA, CDC

C. 2

3. Digital Electronics

The equivalent Boolean algebra expression is: $AB + (\overline{B + C})C$
It simplifies as follows:

$AB + (\overline{B + C})C$
$= AB + (\overline{B} * \overline{C})C$
$= AB + B\overline{C}C$
$= AB + 0$
$= AB$

D. AB

4. Digital Electronics

The equivalent Boolean algebra expression is: $\overline{AB} + (B + C)C$.
It simplifies as follows:
$\overline{AB} + (B + C)C$
$= \overline{A} + \overline{B} + BC + C$
$= \overline{A} + \overline{B} + C$
This is FALSE for (1,1,0) only. Therefore, 7 makes it TRUE.

D. 7

5. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 to get a 3-scoop cones is as follows:

	WB	TR	RB	CB	BA
WB	0	1	1	1	1
TR	0	0	1	1	1
RB	0	0	0	1	0
CB	0	0	1	0	0
BA	0	0	0	1	0

2 =

	WB	TR	RB	CB	BA
WB	0	0	2	3	1
TR	0	0	1	2	0
RB	0	0	1	0	0
CB	0	0	0	1	0
BA	0	0	1	0	0

There are 12 ways to do it:
WB TR RB, WB TR BA (C), WB TR CB, WB CB RB, WB BA CB (D), WB RB CB, TR RB CB, TR BA CB, TR CB RB (A), RB CB RB, CB RB CB, and BA CB RB (B). RB, TR, BA (E) isn't possible.

E. Roasted Beet, Tart Rhubarb, Baked Alaska

6. Graph Theory

The only cycles that exist are of length 2: ABA, CHC. The vertices that are not used are D, E, F, G, which is 4 of them.

B. 4

7. Digital Electronics The equivalent Boolean algebra expression is: $\overline{AB} \oplus ((\overline{B + C})C)$. It simplifies as follows: $\overline{AB} \oplus ((\overline{B + C})C)$ $= \overline{AB} * \overline{(\overline{B + C})C} + \overline{\overline{AB}} * (\overline{B + C})C$ $= \overline{AB} * (B + C + \overline{C}) + AB * \overline{B} * \overline{C} * C$ $= \overline{AB} * 1 + 0$ $= \overline{AB}$ That means there are 6 of them, not (1,1,*).	D. 6
8. Digital Electronics The Boolean algebra expression represented by the following digital circuit is: $\overline{\overline{A} * \overline{(A + B + BC)}}$ $= \overline{\overline{A}} + \overline{A + B + BC}$ $= A + A + B + BC$ $= A + B(1 + C)$ $= A + B * 1$ $= A + B$	D. $A + B$

9. Assembly Language

```
A    DC      7
B    DC      8
     LOAD    B
     SUB     =10
     MULT    =-1
     STORE   C
     LOAD    B
     MULT    C
     STORE   D
     LOAD    A
     MULT    B
     MULT    =100
     ADD     D
     STORE   ANS
     PRINT   ANS
     END
```

This program makes A = 7, B = 8, and C = 10-B = 2. Then D = B*C = 16 so ANS = A*B*100+D = 7*8*100+16 = 5600+16 = 5616. This is also the value of 78 * 72 and the process works for any digits A, B, and C.

E. 5616

10. Assembly Language

```
ANS     DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BL      CONT
        LOAD    ANS
        ADD     =1
        STORE   ANS
CONT    LOAD    N
        STORE   P
        BU      TOP
STOP    PRINT   ANS
        END     ANS
```

This program counts how many "step-downs" there are, in other words, how many times the number is less than or equal to the previous value. These occur when the input is 11, 13, and 7.

D. 3